CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE NAME	
CLASS	**2T**

CHEMISTRY

8872/02

Paper 2

TUESDAY 13 SEPTEMBER 2011
2 Hours

Candidates answer Section A on the Question Paper
Additional Materials: Answer Paper
Data Booklet
Graph Paper (1 sheet)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

SECTION A
Answer all questions.

SECTION B
Answer **two** questions on separate answer paper.

At the end of the examination, fasten all work securely together.
The number of marks is given in brackets [ ] at the end of each question or part of the question.

For Examiner's Use	
Paper 1	**30**
Section A	**40**
Section B	**40**
5 [20]	
6 [20]	
7 [20]	
TOTAL	**110**

This document consists of **15** printed pages and **1** blank page.

[Turn over]

Section A

Answer the questions on the space provided.

1. Chlorine is a powerful oxidant and is commonly used in bleach and disinfectants. As a disinfectant, chlorine compounds are used in swimming pools to keep them clean and sanitary.

Chlorine is able to react with aqueous sodium hydroxide according to the following equation:

$$Cl_2(g) + 2\ NaOH(aq) \longrightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$$

(a) State the type of reaction that occurs when chlorine reacts with aqueous sodium hydroxide. Briefly explain your answer.

Disproportionation.

<u>*Cl*</u> gets <u>oxidized</u> from 0 to +5, and <u>reduced</u> from 0 to -1 simultaneously. [2]

(b) In the above reaction, the sodium chloride formed can be used to produce hydrogen chloride gas by reacting sodium chloride with concentrated sulfuric acid, according to the following equation:

$$NaCl(s) + H_2SO_4(aq) \longrightarrow HCl(g) + NaHSO_4(aq)$$

The hydrogen chloride gas is then bubbled into water, and subsequently diluted in a 250 cm^3 standard flask to form hydrochloric acid, according to the following equation:

$$HCl(g) \xrightarrow{H_2O} HCl(aq)$$

Calculate the volume of chlorine that needs to be reacted with aqueous sodium hydroxide at standard temperature and pressure to produce 0.2 mol dm^{-3} of hydrochloric acid.

No. of moles of HC*l* = $\frac{250}{1000}$ x 0.2 = 0.0500 mol

HC*l* ≡ NaC*l*
No. of moles of NaC*l* = 0.0500 mol

NaC*l* ≡ Cl_2
No. of moles of Cl_2 = 0.0500 mol

Volume of Cl_2 = 0.0500 x 22.4 = <u>1.12 dm^3</u>

[3]

(c) 0.24 g of magnesium ribbon was then added to 100 cm^3 of hydrochloric acid produced in **(b)** in a plastic beaker. The temperature at the start was 20.0 °C, and after the magnesium ribbon has dissolved, it rose to 30.0 °C.

[*specific heat capacity* = 4.18 J g^{-1} K^{-1}].

(i) Give the equation of the reaction between magnesium and hydrochloric acid.

$Mg + 2HCl \rightarrow MgCl_2 + H_2$ ....

(ii) Calculate the enthalpy change of the above reaction.

amount of Mg = 0.24/24.0 = 0.0100 mol
amount of HC*l* = (100/1000)*0.2 = 0.0200 mol

$$\Delta H = \frac{-mc\Delta T}{\text{no. of moles of Mg}}$$
$$= \left(\frac{-(100)(4.18)(10)}{0.01}\right)$$
$$= \underline{-418 \text{ kJ mol}^{-1}}$$

OR

$$\Delta H = 2\left(\frac{-mc\Delta T}{\text{no. of moles of HCl}}\right)$$
$$= 2\left(\frac{-(100)(4.18)(10)}{0.02}\right)$$
$$= \underline{-418 \text{ kJ mol}^{-1}}$$

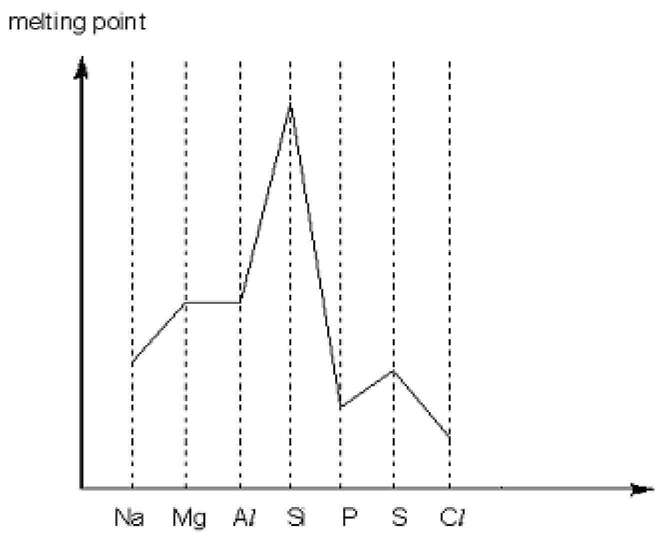
[4]

[Total: 9]

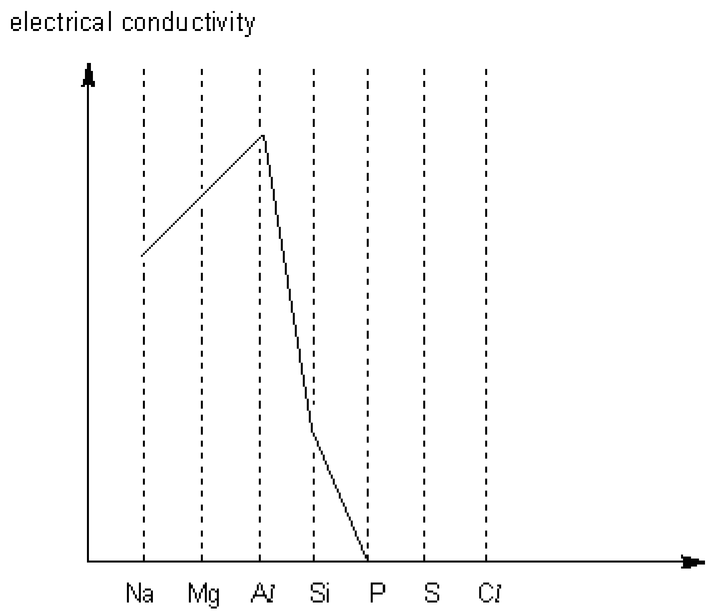
2. Magnesium and aluminium are Period 3 elements in the Periodic Table.

(a) On each of the graphs below, sketch the trends of the properties of the Period 3 elements across the Periodic table.

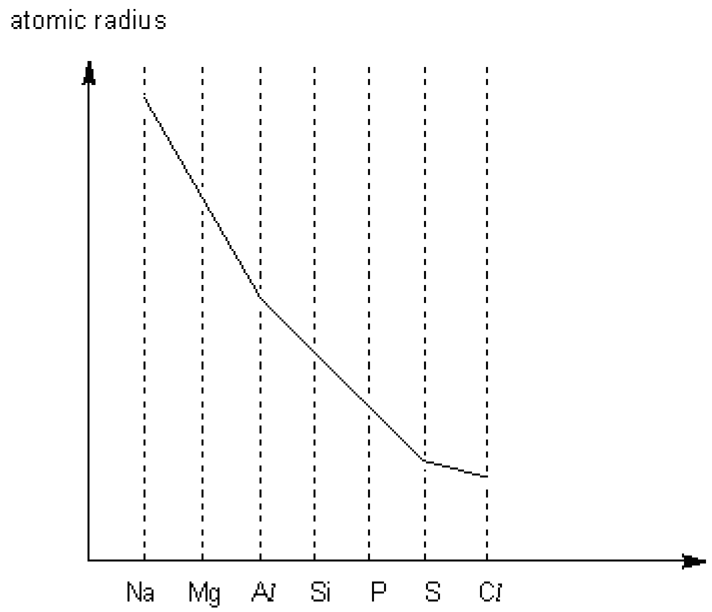
(i) melting point



(ii) electrical conductivity



(iii) atomic radius



[3]

(b) Explain why aluminium has a lower first ionisation energy than magnesium.

Mg: $1s^2 2s^2 2p^6 3s^2$ Al: $1s^2 2s^2 2p^6 3s^2 3p^1$

The first I.E. is lower since it involves the removal of an electron from the <u>3p orbital</u> which is of <u>higher energy</u> / 3p electron is <u>further away</u> from the nucleus [2]

(c) Suggest reasons for the following observations. Write balanced equations for any reactions that occur.

(i) Magnesium oxide is slightly soluble in water but aluminium oxide is insoluble.

Al_2O_3 has <u>stronger ionic bonds/more exothermic lattice energy</u> than MgO, hence, it is insoluble in water as its ionic bonds are too strong to be broken down.

$MgO + H_2O \rightarrow Mg(OH)_2$

(ii) Both magnesium and aluminium chlorides can form acidic solutions in water, but aluminium chloride forms a more acidic solution than magnesium chloride.

Both $MgCl_2$ and $AlCl_3$ can hydrolyse in water to release H^+ ions, hence making the solution acidic.

Al^{3+} has a higher charge density/polarizing power, hence it is able to polarise the water molecules more.

$MgCl_2 + 6H_2O \rightarrow [Mg(H_2O)_6]^{2+} + 2Cl^-$

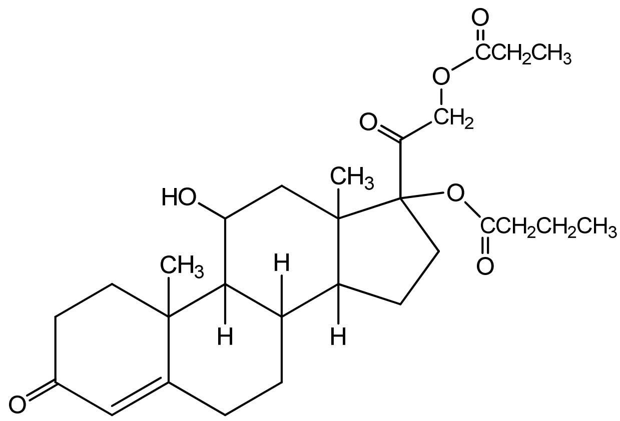
$[Mg(H_2O)]^{2+} \rightarrow [Mg(H_2O)_5OH]^+ + H^+$

$AlCl_3 + 6H_2O \rightarrow [Al(H_2O)_6]^{3+} + 3Cl^-$

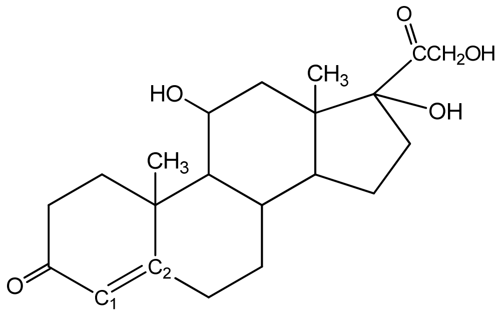
$[Al(H_2O)_6]^{3+} \rightarrow [Al(H_2O)_4(OH)_2]^+ + 2\ H^+$ [6]

[Total: 11]

3. Hydrocortisone and cortisone are types of steriods which are applied to the skin to reduce inflammation, redness and swelling. Their structures are shown below:



Hydrocortisone



Cortisone

(a) With reference to the functional groups present in the structures above, determine which of the two compounds is more soluble in water. Explain your answer.

Cortisone forms <u>more extensive H-bonds</u> with water due to <u>greater number of OH groups</u>. Hence, cortisone is <u>more soluble</u> in water than hydrocortisone.

[2]

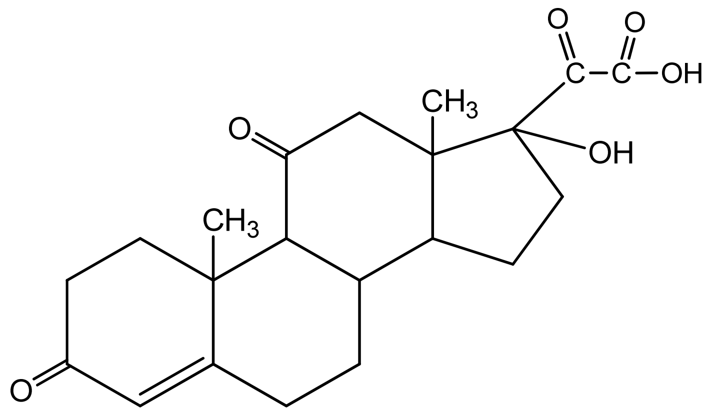
For parts (b) and (c), refer to cortisone only.

(b) (i) Explain why geometric isomerism is **not** possible at carbons C1 and C2.

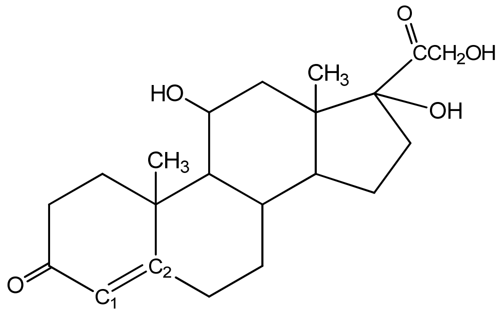
The restricted rotation of C=C is <u>further restricted</u> by a ring which makes trans isomer unlikely to exist.

(ii) Draw the products formed after cortisone undergoes the following reactions:

I. heating with acidified $K_2Cr_2O_7$

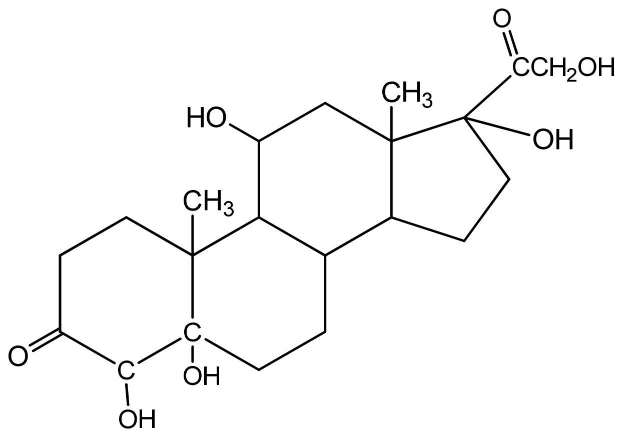


For your convenience, the structure of cortisone is reproduced here.



Cortisone

II. adding $KMnO_4$ at room temperature



[4]

(c) (i) Which functional group(s) present in cortisone reacts with phosphorus(V) chloride?

Functional group(s): ....**alcohol/hydroxyl/-OH group**.................................

(ii) Calculate the mass of PCl_5 required to react with 1 mg of cortisone (M_r = 360).

Cortisone : 3 PCl_5

No. of mol of cortisone = $\frac{1\times10^{-3}}{360}$

= 2.78×10^{-6} mol

No. of mol of PCl_5 = $3 \times 2.78 \times 10^{-6}$ = 8.33×10^{-6} mol

Mass of PCl_5 needed = $8.33 \times 10^{-6} \times 137.5$ = 1.14 mg

[4]

[Total: 10]

4. Carbonated drinks or 'fizzy drinks' (including beer) are best consumed when they are chilled. CO_2 is the gas dissolved in these drinks to give them the 'fizz'. In order for CO_2 to remain in the drink for long periods, the drinks should not be opened. This provides a closed system that allows equilibrium to be established between the CO_2 gas in the air above the drink and the CO_2 dissolved in the drink.

$$CO_2(g) \rightleftharpoons CO_2(aq) \quad \Delta H < 0$$ -------------------------------------- Equilibrium 1

CO_2 is forced into 'fizzy' drinks under considerable pressure as high pressure makes more gas dissolve. When the bottle of drink is opened, a 'pop' or 'fizz' sound is heard. If the drink is not consumed within a few hours after opening, the 'fizz' eventually disappears. At room temperature and pressure, only 90 cm^3 of CO_2 dissolves in every 100 cm^3 of water.

This process of CO_2 dissolving in aqueous solutions occurs everyday naturally, and on a much larger scale. CO_2 dissolves in the oceans such that the gas is taken out of the atmosphere to help maintain a stable environment on earth.

CO_2 is more soluble in water than O_2 or N_2. Hence, much of the excess CO_2 released into the atmosphere from the combustion of fuels is absorbed by the oceans. The cold waters of the Arctic Ocean are expected to absorb more CO_2 than the warmer regions of the earth.

CO_2 molecules are able to react with water to form carbonic acid (H_2CO_3), and the simplified equation that represents this process is:

$$CO_2(g) + H_2O(l) \rightleftharpoons 2\,H^+(aq) + CO_3^{2-}(aq)$$ ----------------------- Equilibrium 2

Near the surface of the oceans, many marine organisms build protective shells composed of insoluble $CaCO_3$ by using the CO_3^{2-} in the seawater.

$$Ca^{2+}(aq) + CO_3^{2-}(aq) \rightleftharpoons CaCO_3(s)$$ -------------------------------- Equilibrium 3

This allows for more CO_2 to dissolve in the oceans. However, despite what is perceived as a greater amount of carbonic acid forming in the oceans, the oceans have been effective in controlling its pH. This is especially important for the oceans to support life. The ocean is thus, considered a buffer solution. The most common type of buffer is made up of a weak acid and one of its salts. The buffering action in the ocean is by carbonic acid and hydrogen carbonate, HCO_3^-.

(a) With reference to Equilibrium 1, use Le Chatelier's principle to explain why

. **(i)** If the drink is not consumed within a few hours after opening, the 'fizz' eventually disappears.

Pressure drops upon opening the drink, and according to LCP, equilibrium 1 will shift <u>left</u> to increase the number of <u>moles of gas</u>. Hence, the CO_2 (aq) that causes the 'fizz' disappears.

(ii) The cold waters of the Arctic Ocean are expected to absorb more CO_2 than the warmer regions of the earth.

In the cold waters of the Arctic ocean, equilibrium 1 lies to the <u>right</u> to favour the forward <u>exothermic</u> reaction. Hence more CO_2 will dissolve in water.

.... [4]

(b) (i) At room temperature and pressure, calculate the amount, in moles, of CO_2 that can dissolve in 500 cm^3 of water.

Volume of CO_2 in 500 cm^3 = 5 x 90 = 450 cm^3

No. of moles of CO_2 = $\frac{450}{24000} = 0.0188$

(ii) At room temperature, an unopened bottle containing 500 ml of carbonated drink has 0.300 moles of CO_2 dissolved in it. How much greater is the pressure in the drink compared to atmospheric pressure?

(Pressure and number of moles of gas are directly proportional.)

Pressure in drink compared to atmospheric pressure = $\frac{0.300}{0.0188} = 16$ **times more**

[2]

(c) (i) Define the term 'buffer solution'.

A buffer solution is a solution that can maintain a fairly constant pH when small amounts of acid or base are added.

(ii) Using a relevant equation, show how the buffering system in the ocean works when it gets too alkaline.

$H_2CO_3 + OH^- \rightarrow HCO_3^- + H_2O$

The OH^- is removed hence pH stays fairly constant. [2]

(d) If the ocean gets too acidic, the following reaction will occur:

$$H^+(aq) + CO_3^{2-}(aq) \rightleftharpoons HCO_3^-(aq)$$

Using the information provided above, suggest and explain what would happen to the protective shells built by marine organisms if the oceans get too acidic.

When the ocean gets too acidic, CO_3^{2-} removes the acid.

When the concentration of CO_3^{2-} drops, equilibrium 3 shifts left and the protective shells composed of insoluble $CaCO_3$ will dissolve.

[2]

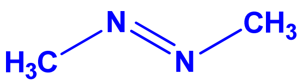
[Total: 10]

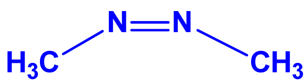
Section B

Answer **two** of the three questions in this section on separate answer papers.

5. Azo compounds are compounds that contain the R-N=N-R' (where R and R' are alkyl groups) functional group. Azo compounds are commonly used as dyes because they have vivid colours like red, yellow and orange. Some of these compounds can also be used as indicators in acid-base titrations. The simplest azo compound is azomethane, $CH_3N{=}NCH_3$.

(a) (i) Draw the structure of azomethane, showing clearly the shape of the molecule. State the bond angles around the carbon and nitrogen atoms.





C: 109°

N: <120°

(ii) With reference to its structure and bonding, comment on

I. the boiling point; and

Azomethane <u>does not have a high boiling point</u> as it has a <u>simple molecular structure</u>and has only <u>weak intermolecular forces</u> that can be broken easily.

II. solubility of azomethane in water. [8]

Azomethane is <u>soluble in water</u> as water molecules can form <u>strong intermolecular forces</u> / <u>hydrogen bonds with the nitrogen atom</u> in azomethane.

(b) When azomethane was heated, it decomposed to form ethane and nitrogen.

(i) Construct an equation for the decomposition of azomethane.

$\mathbf{CH_3N_2CH_3 \rightarrow CH_3CH_3 + N_2}$

(ii) Using relevant bond energy values in the *Data Booklet*, calculate the enthalpy change for the above decomposition reaction.

BE = 2(+305) + 410 - 350 -994

= - 324 kJ mol^{-1}

(iii) Suggest a reason why the **method** in **(b)(ii)** cannot be used to calculate the enthalpy change of formation of sodium chloride.

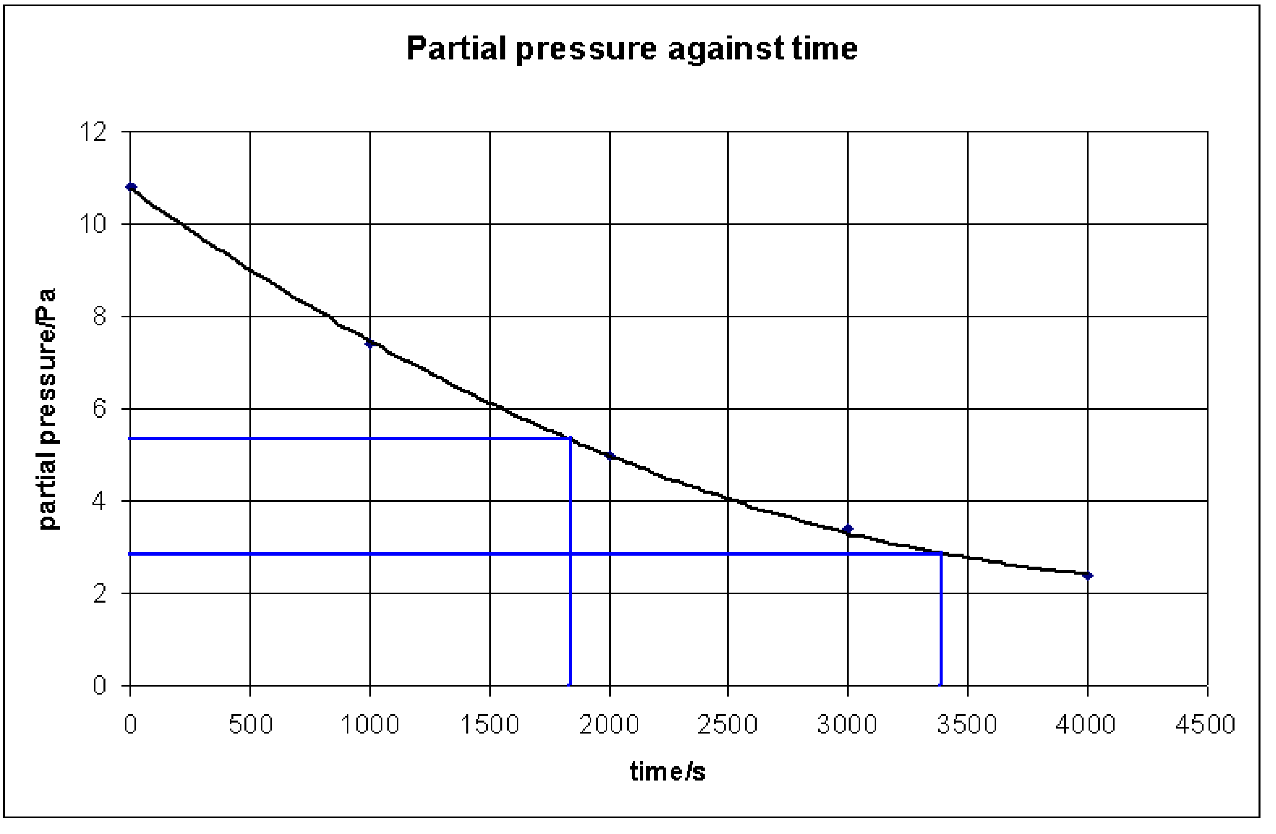
This is because sodium chloride is <u>not a covalent compound</u> / <u>does not have a simple molecular structure</u>.

The *partial pressure* of a gas is the pressure that the gas would exert if it occupies the container alone, and can be taken to be equivalent to its concentration.

The decomposition reaction was followed experimentally by measuring the *partial pressure* of azomethane at various times **t** and the following results were obtained at 600 K.

t/s	partial pressure of azomethane/Pa
0	10.8
1000	7.4
2000	5.0
3000	3.4
4000	2.4

(iv) By plotting a graph of partial pressure against time, show that this decomposition is a first order reaction. Hence, write the rate equation.



Since half-life is constant, this is a first-order reaction.

rate = k[azomethane]

(v) Calculate a value of the rate constant for the reaction.

$t_{1/2}$ = 1500 s

k = $\dfrac{In2}{t_{1/2}}$ **=** $\dfrac{In2}{1500}$ **= <u>4.62 x 10^{-4} s^{-1}</u>**

(vi) Use your graph to determine the partial pressure of azomethane at 1500 s.

partial pressure = 6.1 Pa

(vii) If the initial pressure of azomethane is increased, explain, using *Collision Theory*, the effect on the initial rate of reaction. [12]

When the initial pressure is increases, the number of azomethane molecules increases as well, hence, the frequency of effective collisions increases, leading to an increase in the initial rate of reaction.

[Total: 20]

6. Food digestion involves a chemical process called *hydrolysis*. Proteins, carbohydrates and fats are the three main food chemical groups that can undergo *hydrolysis*.

(a) (i) Explain the term *hydrolysis*.

Hydrolysis is a chemical reaction involving water / reaction in which water is a reagent.

(ii) Using a suitable organic compound of your choice, write a balanced chemical equation to show a *hydrolysis* process. State the reagents and conditions clearly. [3]

Halogenoalkane to alcohol:
R–X (*l*) + NaOH (aq) → R–OH (aq) + NaX (aq)
Aqueous NaOH, reflux.

OR Acid catalysed hydrolysis of nitrile:
$R–CN + 2H_2O + H^+ \rightarrow R–COOH + NH_4^+$
Any mineral acid, reflux

OR Base catalysed hydrolysis of nitrile:
$R–CN + H_2O + OH^- \rightarrow R–COO^- + NH_3$
NaOH or KOH(aq), reflux

OR Acid catalysed hydrolysis of ester:
$RCO_2R' + H_2O \rightarrow RCO_2H + R'OH$
Any mineral acid, reflux

OR Base catalysed hydrolysis of ester:
$RCO_2R' + NaOH \rightarrow RCO_2Na + R'OH$
Aqueous NaOH, reflux.

Fat is an excellent storage medium with an energy content of 9 Kcal g^{-1} (Kcal g^{-1} is kilocalories per gram). This is more than the energy released from carbohydrates (eg. lactose), which is about 4 Kcal g^{-1}. These values represent the amount of energy released during combustion.

(b) (i) Define the standard enthalpy change of combustion.

The standard enthalpy change of combustion, ΔH_c^{o}, is defined as the enthalpy change when one mole of a compound is completely burnt in oxygen under standard conditions of 298K and 1 atm.

(ii) Write a balanced equation, with state symbols, for the enthalpy change of combustion of lactose, $C_{12}H_{22}O_{11}$.

$C_{12}H_{22}O_{11}(s) + 12\ O_2(g) \rightarrow 12\ CO_2(g) + 11\ H_2O(l)$

(iii) Show that the standard enthalpy change of combustion of lactose is -5750 kJ mol^{-1}.

(Given: 1 Kcal = 4.2 kJ and the M_r of lactose is 342)

Energy released in kJ = 4 x 4.2 = 16.8

No. of moles of lactose = $\frac{1}{342}$ = 2.92×10^{-3}

Enthalpy change of combustion $= -\frac{16.8}{2.92\times10^{-3}}$

= - 5750 kJ mol^{-1} (3sf)

(iv) Hence, find the standard enthalpy change of formation of lactose.

(Given: standard enthalpy change of formation of CO_2(g) = -393 kJ mol^{-1}, standard enthalpy change of formation of H_2O(l) = -268 kJ mol^{-1}) [6]

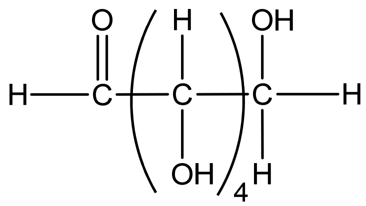
$C_{12}H_{22}O_{11}(s) + 12O_2(g) \rightarrow 12CO_2(g) + 11H_2O(l)$

12(-393) **11(-268)**

elements

Using Hess' Law,
ΔH_f(lactose) + (-5750) = 12(-393) + 11(-268)
ΔH_f(lactose) = -1914 kJ mol^{-1} (or 1910 kJ mol^{-1} to 3sf)

People who have lactose intolerance are unable to digest milk and milk products. These people lack an enzyme called lactase, which is responsible for hydrolysing lactose. Galactose is one of the products of the hydrolysis of lactose. The structure of galactose is shown below:



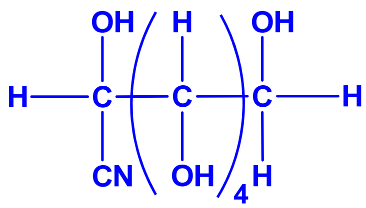
Galactose

(c) (i) Name the functional groups present in galactose.

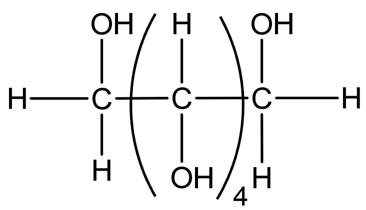
Aldehyde, alcohol

(ii) State the catalyst and draw the organic product formed when galactose reacts with aqueous hydrogen cyanide. [4]

NaCN / KCN / NaOH / KOH.



Galactosemia, a genetic disease, is a result of absence of an enzyme that converts galactose into glucose. In a person with Galactosemia, a toxic compound called dulcitol forms instead. Although not fatal, dulcitol does cause severe stomach upset. The structure of dulcitol is shown below:



Dulcitol

(d) (i) Suggest reagents and conditions that converts galactose into dulcitol in the laboratory. Name the type of reaction occurring.

$LiAlH_4$ in dry ether

OR **$NaBH_4$ in ethanol, heat / reflux**

OR **H_2 with Ni/Pt catalyst, heat.**

Reduction.

(ii) Suggest a simple chemical test to distinguish between galactose and dulcitol. [4]

To separate samples of galactose and dulcitol, add <u>Tollen's reagent and heat</u>. Galactose will give a <u>silver mirror</u> but dulcitol will not give a silver mirror.

OR

To separate samples of galactose and dulcitol, add <u>Fehling's solution and heat</u>. Galactose will give a <u>red precipitate</u> of Cu_2O but dulcitol will not give a red precipitate.

OR
To separate samples of galactose and dulcitol, add <u>2,4-DNPH (Brady's reagent)</u>.

Galactose will give an <u>orange precipitate</u> but dulcitol will not give an orange precipitate.

(e) Fatty acids are classed as saturated, monounsaturated or polyunsaturated, depending on the number of double bonds they contain. The iodine number, expressed in grams of iodine per 100 g of substance, is the amount of iodine that could add to the double bonds. Polyunsaturated fatty acids have two or three double bonds and have an iodine number of 181 and 273 respectively. Suggest the iodine numbers of

(i) Saturated fatty acids

0
There are no double bonds in saturated fatty acids.

(ii) Monounsaturated fatty acids [3]

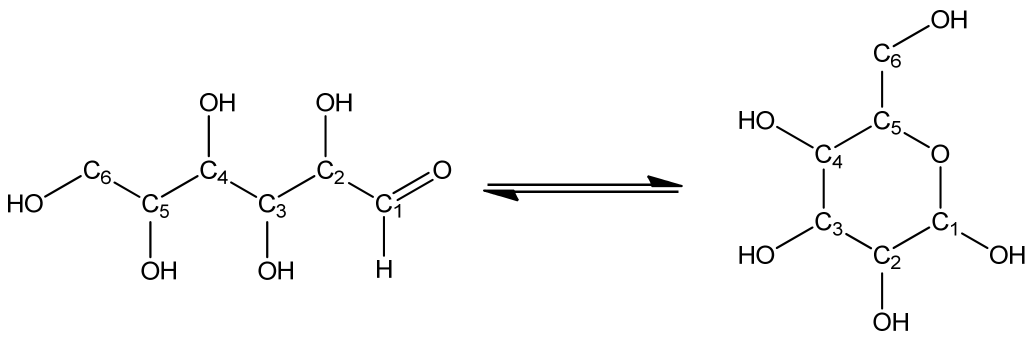
92 (or any value around 80 -100)
It's the difference between two and three double bonds.

[Total: 20]

7. Glucose is referred to as a "reducing sugar" and it is an important carbohydrate in Biology. Cells use it as the primary source of energy.

Glucose exists in different forms, either in chain or cyclic forms, as shown below. The straight-chain form is easily converted into the cyclic form when the -OH on C5 binds to the C1.

[Note: Some hydrogen atoms have been omitted to allow clarity of the structures.]



glucose chain form **glucose cyclic form**

(a) (i) State the type of reaction involved in the conversion above.
Addition

(ii) Given that K_c for the conversion reaction is 5000, state the dominant form of glucose. Hence, find the percentage composition of each form at equilibrium.

Since K_c > 1 (or K_c value is very large), cyclic form is the dominant form.

Let x % be the cyclic form

$$K_c = \frac{x}{(100-x)} = 5000$$

Cyclic form = 99.98 % , chain form = 0.02 %

(iii) In its chain form, glucose reduces Benedict's solution (Fehling's). By considering the change in oxidation number, identify the species that has been reduced. State the functional group in glucose that allows it to be referred to as a "reducing sugar".

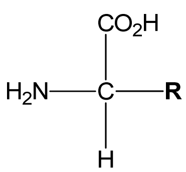
Cu^{2+}(aq) in complex is reduced to Cu_2O(s) – red precipitate

The aldehyde group (-CHO group) is oxidised

[6]

(b) For diabetics who are unable to consume too much sugar, alternative sweeteners are needed. Aspartame is an artificial, non-saccharide sweetener used as a sugar substitute in some foods and beverages.

Hydrolysis of aspartame yields compound **A** shown below:

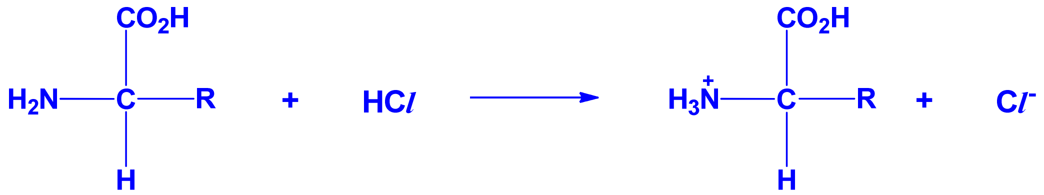


Compound **A**

[Note: **R** represents the rest of the molecule.]

The $–NH_2$ group is considered to be a base, a proton acceptor, due to the availability of a lone pair of electron on N atom. As such, compound **A** can react and dissolve in dilute aqueous HC*l* at room temperature.

(i) Write an equation for the reaction of compound **A** with dilute aqueous HC*l*, and state the type(s) of interactions formed from this reaction.



Dative bond formed between N of $-NH_2$ and H^+/ionic interactions between $-NH_3^+$ and -C $^{-}$

Compound **A** has the following composition by mass: 36.1% carbon, 5.3% hydrogen, 48.1% oxygen, 10.5% nitrogen.

(ii) Find the molecular formula of component **R** in compound **A**.

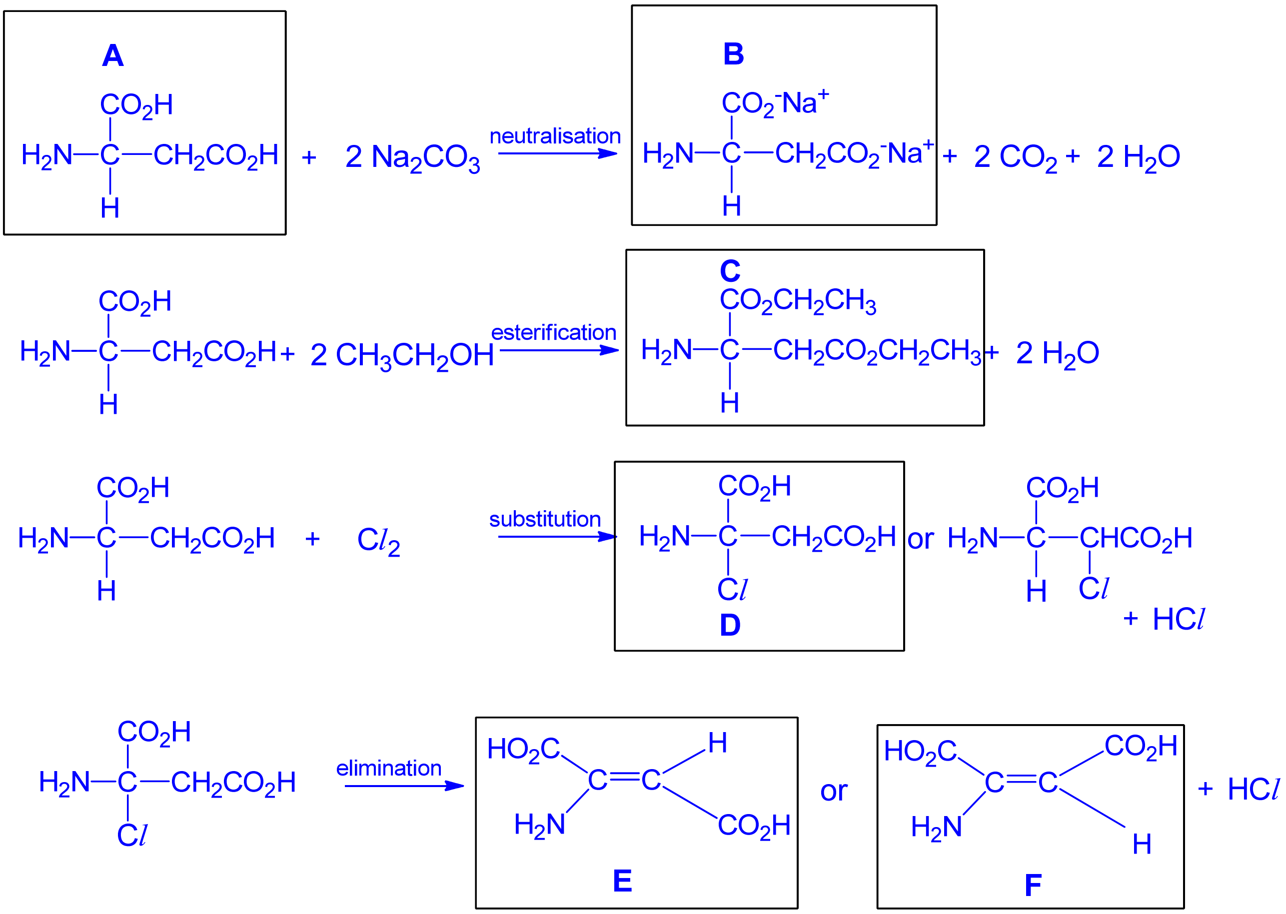
	C	H	O	N
Mol	3	5.3	3	0.75
Ratio	4	7	4	1

Hence, X is $–C_2H_4O_2$.

Compound **A** reacts with 2 moles of Na_2CO_3 to form **B**. **A** also reacts with ethanol in the presence of concentrated sulfuric acid and heat to form **C**. When subjected under uv light and Cl_2(g), **A** forms a monochlorinated compound, **D**. When **D** reacts with ethanolic NaOH and heated, isomeric compounds **E** and **F** are formed.

(iii) Identify the structural formulae of compounds **A** – **F**, explaining the chemistry involved and writing equations where applicable.

A – <u>have 2 acidic groups</u>. Hence, there is $–CO_2H$ group in X.


A
B
neutralisation
+ 2 Na_2CO_3
+ 2 CO_2 + 2 H_2O
C
esterification
+ 2 CH_3CH_2OH
+ 2 H_2O
substitution
+ Cl_2
D
or
+ HCl
elimination
E
or
F
+ HCl

[14]

[Total: 20]